Exelon Generation.

Clinton Power Station
8401 Power Road
Clinton, IL 61727

U-604340
April 21, 2017

10CFR50.73
SRRS 5A.108

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2017-001-00

Enclosed is Licensee Event Report (LER) 2017-001-00: Failure of the 138 kV Offsite Power Source Results in a Loss of Secondary Containment Vacuum. This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. Dale Shelton, Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Theodore R. Stoner
Site Vice President
Clinton Power Station

KP/cac

Attachment: Licensee Event Report 2017-001-00

cc:

Regional Administrator— NRC Region III
NRC Senior Resident Inspector - Clinton Power Station
Office of Nuclear Facility Safety — Illinois Emergency Management Agency

IE22
NRR

NRC FORM 366 (11-2015) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
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4. TITLE

Failure of the 138 kV Offsite Power Source Results in a Loss of Secondary Containment Vacuum

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER 05000
02	24	2017	2017	- 001 -	00	04	21	2017	FACILITY NAME	DOCKET NUMBER 05000

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
099	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Dale A. Shelton, Regulatory Assurance Manager	217-937-2800

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☒ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On February 24, 2017 at approximately 2239 hours (CDT), the Main Control Room (MCR) received numerous annunciators which indicated a loss of the 138 kV off-site supply to the Emergency Reserve Auxiliary Transformer (ERAT). The MCR entered Technical Specification (TS) Limiting Condition for Operation (LCO) 3.8.1, AC Sources-Operating, Required Actions A.1 and A.2. The Division 1 Fuel Building ventilation (VF) system isolation dampers closed due to the voltage transient causing a trip of the VF supply and exhaust fans and Secondary Containment (SC) vacuum to rise to slightly greater than 0 inches water gauge (WG), exceeding the TS requirement of greater than 0.25 inches vacuum WG. The MCR entered Emergency Operating Procedure (EOP)-8, Secondary Containment Control, and TS LCO 3.6.4.1, Secondary Containment, Required Action A.1. SC vacuum was restored within TS requirements at approximately 2242 hours by starting the Standby Gas Treatment (VG) system. The Transmission System Operator confirmed that the loss of the 138 kV line was due to a line fault external to the Station. The 138 kV line was successfully re-energized at 0053 hours on February 25, 2017 and the ERAT off-site source was restored and declared OPERABLE at 0300 hours. This event is reportable per 10CFR50.73(a)(2)(v)(C).

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric—Boiling Water Reactor, 3473 Megawatts Thermal Rated Core Power
Energy Industry Identification System (EIIS) codes are identified in the text as [XX]

EVENT IDENTIFICATION

Failure of the 138 kV Offsite Power Source Results in a Loss of Secondary Containment Vacuum

A. Plant Operating Conditions before the Event

Unit: 1	Event Date: 02/24/17	Event Time: 2239
Mode: 1	Mode Name: Power Operation	Reactor Power: 99 percent

B. DESCRIPTION OF EVENT

On February 24, 2017, at approximately 2239 hours CDT the Emergency Reserve Auxiliary Transformer (ERAT) and the associated Static VAR Compensator (SVC) tripped due to a loss of the 138 kV line owned and maintained by Ameren Illinois. The Main Control Room (MCR) entered Technical Specification (TS) Limiting Condition for Operation (LCO) 3.8.1, AC Sources-Operating, Required Actions A.1 and A.2. The Clinton area was experiencing very strong winds at the time of the 138 kV power loss and the ERAT trip.

As a result of the voltage transient caused by automatic transfer to the Reserve Auxiliary Transformer (RAT), the Division 1 Fuel Building Ventilation (VF) system isolation dampers closed causing a trip of the VF supply and exhaust fans. The effect of this condition was that Secondary Containment (SC) vacuum rose to slightly greater than 0 inches water gauge (WG) and exceeded the TS limit of -0.25 inch WG. Operations personnel entered Emergency Operating Procedure (EOP) -8, Secondary Containment Control, and TS LCO 3.6.4.1, Secondary Containment, Required Action A.1. As a result of the loss of power and subsequent bus transfer, the following additional system actuations occurred:

Division 1 MCR Ventilation Chiller tripped
Division 1 Diesel Generator room Exhaust Fan 1VD02CA tripped,
Division 1 Shutdown Service Water Pump auto started,
Fission Product Monitoring (FPM) skid isolation valves shut,
Division 1 Hydrogen Monitoring system isolations valves shut,
Containment Service Air system Outboard Isolation 1SA029 shut
Nuclear Systems Protection System Inverter Trouble alarms annunciation (due to loss of syncronization).

NARRATIVE

The system actuations were addressed and plant equipment was restored to service in accordance with plant procedures. The Standby Gas Treatment System (VG) was manually started and SC vacuum was restored within TS limits at approximately 2242 hours. The momentary loss of power did not adversely affect the safe operation of the plant or the restoration of SC.

During the investigation, Ameren determined that high winds caused a line fault external to the Station which resulted in the 138 kV supply line trip. At 0053 hours on February 25, 2017 Ameren successfully re-energized the 138 kV line. The ERAT was returned to service and declared OPERABLE at 0300 hours.

C. CAUSE OF EVENT

The cause of the event was identified by Ameren as an air flow spoiler failure. The radial feed design of 138 kV offsite power to CPS is susceptible to single point failures. The Clinton area was experiencing very strong winds at the time of the air flow spoiler failure.

D. SAFETY ANALYSIS

This event is reportable in accordance with 10 CFR 50.73(a)(2)(v)(C) as a condition that could have prevented fulfillment of a safety function to control the release of radioactive material because SC was declared inoperable during the event.

The broken air flow spoiler which caused a trip of the 138 kV supply line and the ERAT is a condition bounded by the analysis in Updated Safety Analysis Report section 15.2.6, "Loss of AC Power." The condition described in this report is less severe and involved no safety consequences. The AC electrical power sources are designed with sufficient redundancy to ensure the availability of necessary power to plant systems, structures, and components.

The VG system was fully operable at the time of the event and capable of performing its required safety function. Operations entered EOP-8 for SC vacuum less than -0.25 inch WG and entered TS LCO 3.6.4.1 Required Action A.1. SC vacuum was restored to within TS limits within the associated completion time requirement. The SC vacuum is kept slightly negative relative to atmospheric pressure to prevent leakage to the atmosphere. The VF system is a non-safety ventilation system which is normally in service to maintain SC vacuum. The VG system is the safety-related system which is relied upon to perform this function following an accident. During the event the VG system was manually placed in service and restored SC to operability consistent with its safety function. Therefore, the ability of the station to maintain SC vacuum in the event of an accident was never jeopardized or challenged by the VF system fan trip. Engineering analysis has determined that this event is not considered a safety system functional failure.

The event discussed in this report does not constitute a safety system functional failure.

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E. CORRECTIVE ACTIONS

The 138 kV line was successfully re-energized on February 25, 2017 at 0053 hours. Following the restoration of the 138 kV line, the ERAT transformer and associated static VAR Compensator were returned to service. An additional corrective action includes installing a modification to preclude tripping of VF due to a momentary loss of power.

F. PREVIOUS SIMILAR OCCURENCES

There were no previous occurrences involving the loss of 138 kV line and ERAT trip due to a broken air flow spoiler. However, several similar occurrences involving to loss of the 138 kV line have occurred. These events are summarized below.

License Event Report 2016-004-00: Trip of Emergency Reserve Auxiliary Transformer Static VAR Compensator Causes Positive Secondary Containment Pressure Following Lightning Strike on 138 kV Offsite Source

On March 30, 2016, at 1545 CDT, the Emergency Reserve Auxiliary Transformer (ERAT) Static VAR Compensator (SVC) [COMP] tripped when the 138 kV line cycled open and closed due to a lightning strike. During the event, Division 1 Secondary Containment (SC) isolation dampers went shut due to the momentary loss of power and SC vacuum exceed 0 inches water gauge (WG). EOP-8 and TS LCO 3.6.4.1, Required Action A.1 were entered as a result of this abnormal condition. At 1547, the Standby Gas Treatment System (VG) Train A was started per CPS procedures. At 1550, SC vacuum had been restored to TS limits. Procedure EOP-8 and TS LCO 3.6.4.1 Required Action A.1 were exited. The ERAT was restored at 1926 hours and TS LCO 3.8.1 Required Actions A.1 and A.2 were exited.

Licensee Event Report 2015-004-00: Trip of Emergency Reserve Auxiliary Transformer Static VAR Compensator Causes Positive Secondary Containment Pressure Following Lightning Strike on 138 kV Offsite Source.

On June 25, 2015 at 0301 CDT, the Main Control Room received numerous annunciators that indicated a trip of the ERAT transformer Static VAR Compensator (SVC)[COMP] caused by a voltage transient on the 138 kV offsite source due to thunderstorms in the area. The Division 1 Safety Bus [BU] was manually aligned from the reserve source to its normal source. As a result of the voltage transient, the Division 1 VF system isolation dampers closed causing a trip of VF supply and exhaust fans. SC vacuum increased to greater than 0 inches WG which exceeded the TS requirement. SC vacuum was restored within TS requirements by reopening the VF isolation dampers and restarting the VF supply and exhaust fans. The ERAT SVC was returned to service at 0457 CDT.

NARRATIVE

G. COMPONENT FAILURE DATA

The component failure which caused the 138 kV supply trip was owned by Ameren. There were no CPS component failures associated with this event.